

INDIAN INSTITUTE OF TECHNOLOGY KHARAGPUR

1. Name of the Academic Unit: Computer Science and Engineering

2. Subject Name: Database Management Systems L-T-P: 3-0-0 Credits: 3

3. Pre-requisites: Discrete Structures (CS21201)

4. Syllabus and reference books:

Syllabus:

Database system architecture: Data Abstraction, Data Independence, Data Definition and Data Manipulation Languages. Data models: Entity-relationship, relational and object oriented data models, integrity constraints and data manipulation operations. Relational query languages: Relational algebra, tuple and domain relational calculus, SQL. Relational database design: Domain and data dependency, Armstrong axioms, normal forms, dependency preservation, lossless design. Query processing and optimization: Evaluation of relational algebra expressions, query equivalence, join strategies, query optimization algorithms. Storage strategies: Indices, B-trees, hashing. Transaction processing: Recovery and concurrency control, locking and timestamp based schedulers, multiversion and optimistic concurrency control schemes. Advanced topics: Object-oriented and object relational databases, logical databases, web databases, distributed databases, data warehousing and data mining.

Reference Books:

- 1) Abraham Silberschatz, Henry Korth, and S. Sudarshan, Database System Concepts, McGraw Hill.
- 2) Raghu Ramakrishnan, Database Management Systems, WCB/McGraw Hill.
- 3) Bipin Desai, An Introduction to Database Systems, Galgotia.
- 4) J. D. Ullman, Principles of Database Systems, Galgotia.
- 5) R. Elmasri and S. Navathe, Fundamentals of Database Systems, Addison-Wesley.
- 6) Serge Abiteboul, Richard Hull and Victor Vianu, Foundations of Databases. Addison-Wesley.

5. Lecture-wise break-up:

Sl. No.	Topic	No. of lectures
1.	Database system architecture: Data Abstraction, Data Independence, Data Definition and Data Manipulation Languages.	3
2.	Data models: Entity-relationship, relational and object oriented data models, integrity constraints and data manipulation operations.	6
3.	Relational query languages: Relational algebra,	6

	tuple and domain relational calculus, SQL	
4.	Relational database design: Domain and data dependency, Armstrongs axioms, normal forms, dependency preservation, lossless design.	6
5.	Query processing and optimization: Evaluation of relational algebra expressions, query equivalence, join strategies, query optimization algorithms.	3
6.	Storage strategies: Indices, B-trees, hashing.	7
7.	Transaction processing: Recovery and concurrency control, locking and timestamp based schedulers, multiversion and optimistic Concurrency Control schemes.	6
8.	Advanced topics: Object-oriented and object relational databases, logical databases, web databases, distributed databases, data warehousing and data mining.	3
Total number of hours		40